

Understanding the Dynamics in Creating Domain-Specific AI Design Guidelines: A Case Study of a Leading Digital Finance Company in South Korea

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ABSTRACT

Artificial intelligence (AI) significantly impacts the user experience (UX) across a wide range of products and services. However, many practitioners find it challenging to design the UX for AI-enhanced products. Although there are human-AI interaction design guidelines available, they are often too generic to fully address domain-specific challenges in fields such as finance and healthcare. Additionally, they sometimes overlook the constraints imposed by different working environments. In this research, we conducted a case study at one of the leading financial corporations in South Korea to develop their own AI product interaction guidelines. We interviewed 13 practitioners including bank UX practitioners, design agency UX designers, and academic UX researchers who were involved in this initiative, to gather insights into their experiences with crafting these guidelines. Our findings revealed a mixed reception towards existing AI design resources among practitioners, underscoring the necessity to incorporate elements specific to banking services, such as the company's customer experience (CX) principles, overarching agent personas, multifaceted channels, and diverse interaction modalities. We also observed the importance of understanding the dynamics among various stakeholders within a company. We discuss the implications and design considerations for creating domain-specific resources specialized in AI interactions.

CCS CONCEPTS

• **Human-centered computing** → **User centered design**; *Human computer interaction (HCI)*.

KEYWORDS

AI, Interaction Design, Guidelines, UX, CX, Human-AI Interaction

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1 INTRODUCTION

Artificial intelligence (AI) is enhancing the user experience across a variety of products and services. The emergence of cutting-edge algorithms has unlocked unprecedented possibilities in AI products and services, offering highly efficient, enjoyable, and personalized experiences. Academia and industry are paying close attention to the potential of AI, striving to create a range of innovative products and services [24]. This trend is further fueled by recent rapid development and growing interest in generative AI. Companies across sectors such as finance [6], healthcare [29], law [19], transportation [17], and gaming [15] are eagerly adopting various forms of AI, such as LLMs and generative AI, to stay ahead of (or at least not fall behind) their competitors.

However, UX practitioners, who are at the forefront of designing these AI products and services, face significant challenges [26]. Despite the many advantages of AI, it can be unpredictable or complex, surpassing the capabilities of UX practitioners accustomed to designing static and predictable products and services [5, 25, 26]. Moreover, most UX practitioners, lacking expertise in data science or machine learning, struggle to understand what AI can and cannot do. This problem is intensifying with the increasing emergence of more complex algorithms, including generative AI. Consequently, there is a growing need for resources to help UX practitioners envision AI capabilities and leverage them to design innovative UX products and services.

As part of these efforts, several tech giants and researchers have published resources for AI product design [1, 2, 9, 10, 13, 18]. Notable examples include Microsoft's Guidelines for Human-AI Interaction [1], Google's People + AI Guidebook [9], and Apple's Human Interface Guidelines [2]. These guidelines focus on several key issues surrounding AI, such as explainability, user control, and error feedback. They also consider the practical application of AI,

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offering principles, use cases, and in some instances, worksheet formats. In this regard, Yildirim et al. qualitatively explored how these guidelines are utilized in the actual work of UX designers [28]. The study found that these guidelines are primarily used for educational purposes, problem framing, and facilitating a hybrid approach between user-centric and AI-centric design. While this research offers valuable insights for generalizing UX practices, it also raises questions about its real-world applications in specific sectors (such as finance, law, or mobility), different cultural contexts (small/large business or company values), and stakeholder dynamics.

In this research, we conducted a case study on the dynamics of creating AI design guidelines within a company using AI for a specific domain. We recruited various stakeholders from a South Korean financial group's AI interaction design guideline project and conducted qualitative research on their experiences and the resulting outcomes. We conducted in-depth interviews with 13 participants comprising bank practitioners, designers from a design agency, and academic researchers from a university. We inquired about the reasons behind the project, the role of existing guidelines for large tech companies, and considerations for AI guidelines in the finance sector. It is essential to clarify that our research focus is not on the outcomes themselves, but on understanding the challenges practitioners face in creating AI design guidelines and identifying potential areas for improvement, which we intend to introduce new perspectives to areas of AI application. We observed a mixed reception towards existing AI design resources among practitioners, underscoring the necessity to incorporate elements specific to banking services, such as customer experience (CX) principles, agent personas, multifaceted channels, and diverse interaction modalities. One significant finding was the essential role of understanding the dynamics among various stakeholders within a company, which can profoundly influence the guideline development process. Based on these findings, we propose implications and considerations for designing AI guidelines in specific domains, particularly finance.

This research contributes in two ways: (1) by presenting empirical results on the role of AI guidelines in the finance sector and their significant features, and (2) by suggesting how AI guidelines should be tailored for specific sectors, and their roles in complex organizational structures and communication processes.

2 RELATED WORKS

Recent advancements in AI have sparked significant interest in how it can be integrated into the design of products and services, particularly in the fields of UX and human-computer interaction (HCI). The explosive growth of generative AI technologies has especially drawn heightened attention from academia and industry.

However, the emergence of these new technologies presents various challenges for UX practitioners [26]. Designers often find it difficult to understand the capabilities and limitations of AI, which hampers their creative and design processes [5, 25]. They struggle to envision novel, implementable AI solutions during the early stages of design [27] and face difficulties in iteratively prototyping and testing human-AI interactions [22]. Setting appropriate user expectations for the sometimes unpredictable outputs of AI is another challenge [12], as is collaborating effectively with AI

engineers due to a lack of shared workflow, boundary objects, or a common language [7, 14, 25].

Subsequently, research has focused on developing resources for UX practitioners to use in AI design [2, 8, 10, 13, 18]. A prime example is Microsoft's Guidelines for Human-AI Interaction [1]. These guidelines are designed to enhance human-AI interaction, ensuring that AI systems are technically robust and can interact with users intuitively, reliably, and ethically. Another notable resource is Google's People + AI Guidebook [9], a collection of best practices and recommendations to aid in building human-centered AI products. It guides teams in creating AI applications that are advanced yet easy and enjoyable to use, emphasizing the human aspect of AI interactions. This resource includes various cases, worksheets, and a section for beginners. Recent research on AI guideline usage in practice shows that practitioners use them not only to address AI design challenges but also for education, internal resource development, cross-functional alignment, and gaining trust and acceptance within their teams and organizations [16, 28].

Some researchers have proposed more active forms of resources for AI design, such as checklists [11, 18, 20] or design patterns [21, 23]. Madaio et al. created a co-designing checklist to explain the organizational challenges and opportunities regarding fairness in AI [18]. Their iterative co-designing process with UX practitioners revealed that AI fairness checklists could provide an organizational infrastructure to formalize ad-hoc processes and empower individual advocates, and discuss how organizational culture might impact the effectiveness of such checklists.

Despite the overall usefulness of these various resources, they have been criticized for not fully encompassing the specific and diverse issues inherent in AI. Particularly, these resources, often developed for universal use by large corporate research labs, may not directly apply to smaller businesses, unique work environments across different cultures, or specialized fields like finance, law, or healthcare. Moreover, they may be unable to keep pace with the rapidly evolving and changing technological considerations of AI, such as LLMs, generative AI, and conversational agents.

3 METHODOLOGY

To understand how AI design guidelines are created in specific domains, and the dynamics involved in the process, we conducted a detailed case study by selecting a particular AI guideline example project in our company and contacting members of the project team. It is important to note that this paper was conducted after the completion of the project, ensuring that our research did not influence the project's outcomes. Additionally, our study does not describe the project's final product (the AI design guidelines) itself; rather, the aim is to understand the design processes and considerations involved, and to present the implications drawn from these aspects.

3.1 AI Interaction Design Guidelines for a Banking Company

The primary focus of our research was to establish AI interaction guidelines for a bank. The bank we selected is one of the leading financial institutions in South Korea and is well-recognized for

providing a wide array of financial services including banking, investments, and insurance. The company has been integrating AI into its various services, showing significant interest in innovating CX. The bank's AI services include four service channels: AI Banker, a kiosk with a virtual AI avatar to guide users in bank branches; mobile banking services, voice ARS service, and chatbots in its mobile application. These services interact with customers through various forms of interaction modalities (text, sound, touch, etc.) in diverse user interfaces and applications. To design a cohesive AI UX across its various services, the company sought to create systematic and specialized AI design guidelines. The project involved multiple stakeholders including a design agency and UX researchers from an information school of a local university. The design agency was responsible for the detailed design of UI components in the guidelines. At the same time, the university researchers handled extensive literature reviews, case analyses, customer interviews, AHP analysis [4], design workshops, and heuristic evaluations, focusing on the main structure and high-level principles of the AI interaction guidelines. The project took about six months to complete, with regular weekly and monthly collaborative meetings among all stakeholders to produce the final output. This collaborative effort led to the successful creation of the AI interaction guidelines.

The AI interaction guideline structure is broadly composed of AI personas, AI principles, and detailed guidelines for each of the four channels. AI personas define the top characteristics that the bank's AI services should embody, in three main features. The AI principles consist of ten high-level principles based on the overall CX principles of the bank. The detailed AI guidelines provide specific explanations and detailed descriptions of design components and use cases for each of the four channels. The elements of these AI guidelines are hierarchically organized and logically interconnected.

3.2 Participants

For our study, we aimed to recruit various stakeholders who participated in the AI interaction design guidelines project. These participants can be broadly categorized into three institutional groups: the bank (client), the design agency, and the university research team. Initially, we recruited three participants from the bank, the client for this project. They included UX practitioners from a team called the CX Innovation Center. Next, we recruited four individuals from the design agency. These participants were responsible for the overall project management, including scheduling and detailed design of UI components. Lastly, the university research team consisted of graduate students specializing in UX and HCI. These students participated in preliminary research, user surveys, the derivation of AI principles, and heuristic evaluations. We recruited six members from this group. During the six-month project, these participants regularly met for collaboration, developing positive working relationships (details of the participants are shown in Table 1).

3.3 Process

We conducted in-depth interviews with these participants. The interviews began one month after the completion of the project and lasted approximately two weeks. On average, each interview took

about 40 minutes. About half of the participants were interviewed in person, while the rest were interviewed via Zoom video conferencing based on their preferences. At the start of each interview, the research team briefly explained the purpose and procedure of the interview and then obtained consent forms signed by the participants. The interviews were structured in three stages. The first stage involved asking participants for detailed descriptions of their roles and tasks in the project. The second stage requested their opinions on the important aspects of the project's execution and detailed evaluations of the deliverables produced. Finally, the third stage involved general questions about their experiences during the project, including challenges, positive aspects, and areas for improvement. Participants were allowed to use web browsers during the interview to refer to related examples and resources (including other AI guidelines and cases), and they could also refer to the final AI interaction guidelines while answering. As compensation for their participation, each interviewee received a \$10 gift card.

3.4 Analysis

All interviews were recorded, and we transcribed every recorded file. We conducted a thematic analysis of the transcribed scripts [3]. First, to familiarize themselves with the interview outcomes, all research team members thoroughly reviewed each line of the transcribed scripts. Following this, we marked the important parts of each line and added appropriate summaries or keywords through a coding process. We then aimed to derive higher-level themes centered around these codes. All team members reviewed these themes again and returned to the scripts for further review. This iterative process was repeated until all researchers reached a consensus, allowing us to ultimately identify three major themes. We will describe these themes in the Findings section below.

4 FINDINGS

We identified three major findings regarding (a) practitioners' evaluation of existing AI design resources, (b) considerations for integrating AI into the field of finance, and (c) the importance of understanding the dynamics within the company.

4.1 Practitioners' Evaluation of Existing AI Guideline Resources

Our first notable finding was the mixed reactions of UX practitioners toward the widely recognized AI design resources. Key references for many involved in this project were Microsoft's Guidelines for Human-AI Interaction [1] and Google PAIR's People + AI Guidebook [9]. Google's Material Design [8] was also extensively consulted for UI component design. Although these resources were widely respected for their quality, participants also highlighted several drawbacks.

Participants praised these resources for their credibility and practicality as guidelines in AI product and service design. Google Material Design was particularly valued, often seen as a foundational guideline in mobile app and web service design. P2 remarked, "*Material Design is incredibly specific and almost like a textbook for designers. It's the basis for many companies creating their own design systems, and it's a go-to resource for clarification in general UX quandaries.*" P1 added, "*It instills trust, covering a wide array*

Table 1: Participants Information

Participant ID	Team	Role	Age	Gender	Experience
P1	Bank (Client)	UX practitioner	24	F	2 years
P2	Bank (Client)	UX practitioner	28	F	2 years
P3	Bank (Client)	UX practitioner	40	M	14 years
P4	Design agency	Project manager	49	M	22 years
P5	Design agency	UI/UX designer	39	M	9 years
P6	Design agency	UI/UX designer	38	M	8 years
P7	Design agency	Product designer	41	M	14 years
P8	University research team	UX researcher	30	F	5 years
P9	University research team	UX researcher	30	F	5 years
P10	University research team	UX researcher	26	M	2 years
P11	University research team	UX researcher	28	F	4 years
P12	University research team	UX researcher	26	F	2 years
P13	University research team	UX researcher	25	F	2 years

of components. Being a standard set by Google, it lends a degree of credibility and assures that following its design principles is unlikely to lead to major UX issues.”

Microsoft’s Guidelines for Human-AI Interaction was praised for addressing AI-specific issues across various sectors and offering customized advice for designing AI products and services. Their user-friendly format, which organizes principles according to user interaction stages in an easily accessible card layout, was highly valued. P3 observed, “Microsoft’s guidelines are intuitive and impressive, categorizing interactions into stages like initially, during interaction, when wrong, and over time.” Similarly, the Google PAIR Guidebook was consistently referenced for AI-specific projects. It was seen as more hands-on than Microsoft’s guidelines, covering a wider range of AI-specific issues including real-world application examples, practical worksheets, and supplementary materials like glossaries and blog articles. P2 commented, “The Guidebook is great with clearly defined cases and even workshop kits.” P11 noted, “The guidelines are presented along with various cases from Google’s services.”

These tools were utilized not only as guides for designing tangible products and services, but also as educational materials for internal training and departmental communication, resonating with Yildirim et al.’s research findings [28]. P1 mentioned, “They talk about universally applicable principles and patterns, which are excellent for enhancing UX understanding, particularly for educational and study purposes when starting AI services.” Nevertheless, the participants identified some limitations of these resources. They criticized them for being somewhat general and broad, which posed difficulties in specific practical applications. For example, P3 expressed, “I felt they were quite general, almost like basic guidelines from the past. So, there are many constraints when it comes to practical use.” A particular concern was the lack of representation of sectors like finance, in which the participants were closely involved. The resources were also seen as inadequate in reflecting unique company circumstances such as corporate identity and complex stakeholder dynamics. Acknowledging their usefulness, the practitioners emphasized the need for these resources to address specific and detailed AI-specialized issues more comprehensively. P10

stated, “They are a good starting point or a benchmark, but they don’t perfectly fit the finance sector.”

Additionally, they highlighted that AI services encompass a range of modalities like text, voice, touch, and various formats like chatbots and agent kiosks. However, they pointed out that these resources often adhere to traditional principles of mobile or web UI, which was a major area of criticism. P6 said, “They might be useful for mobile or web UI design, but they fall short in proposing new interaction methods that come to mind when thinking of AI.” Moreover, while they deeply empathized with the importance of emphasizing ethical considerations in these resources, they also noted that such aspects often seem to be overlooked in practice. P9 mentioned, “These resources cover not just practical aspects but also ethical considerations. However, in our guidelines, that aspect wasn’t given much weight.”

4.2 Important Factors in Integrating AI in the Field of Finance

Our inquiry into what participants deemed more important than existing resources revealed a deep contemplation of the nature of the finance and banking sector. P7 reflected, “In my view, the most meaningful and beneficial aspect of our project was creating AI guidelines tailored specifically for the finance domain.” This project aimed to encapsulate the bank company’s CX values, essentially the company’s overarching vision. To achieve this, the university research team conducted initial focus groups with current bank customers to explore the bank’s perceived image, the values customers expect, and customer expectations for the bank’s AI services. P2 mentioned, “Since our guidelines are for our bank’s AI interactions, they needed to be custom-made to align perfectly with the values and vision of the CX we aim for in the future.”

This process led to the establishment of primary CX principles including trustworthiness, accuracy, a consistent and integrated experience, systematic design, user-centricity, security, ease of accessibility, simple and intuitive design, acceptance of new technologies, and adherence to social values. P12 observed, “Our guidelines included top-level principles. I believe these provide a solid foundation for application in a financial context. We shaped the guidelines around

these values.” These principles reflect the conservative standards necessary in finance, where customers must feel secure entrusting their assets, requiring extra caution in products or services to avoid potential errors or mistakes. The need for a consistent and integrated experience across various banking operations such as savings, loans, foreign exchange, and investments was also emphasized. P6 explained, *“I aimed to create guidelines that specify how certain financial activities like transfers, withdrawals, inquiries, and loans should be handled when applied to AI.”*

In addition to these principles, the personas of AI agents were considered vital. Given the nature of AI, personified agents could be present in both tangible and intangible forms throughout the service. It was essential to establish common personas for these agents. The university research team developed representative characteristics of these personas through user surveys, CX philosophy and principles, and literature reviews. P12 shared, *“While creating these guidelines, we thought it would be beneficial if the unique characteristics of agents for various channels were highlighted appropriately, leading to the development of multiple agent personas.”* These personas included being (a) trustworthy and reliable, (b) quick and confident, and (c) friendly and kind. The personas were designed to serve as mediums for communication, bases for design decisions, or guidelines during the AI product design process (P5, P12).

Finally, participants emphasized the importance of considering various service channels and interaction modalities. P7 stated, *“It was crucial to consider how the offline experience transitions to digital.”* In this context, key channels included chatbots, voice ARS, and AI bankers in kiosks. These channels involve diverse modalities of input and output such as text, voice, and touch. P10 noted, *“With the introduction of digital desks and AI bankers, people aren’t limited to a single modality of interaction, so a multifaceted approach was necessary.”* For example, chatbots assist users in performing banking tasks via a conversational UI, where users can issue commands through text or button clicks, and the chatbot responds with text or visual information such as infographics. Voice input involves interpreting vocal commands and providing feedback. P6 pointed out, *“For instance, if I say my password to an AI, someone nearby might overhear. In such cases, users should be prompted to input their password on the screen through touch or other interaction forms.”* For AI bankers, interacting in human form on kiosks required careful design of aspects like facial expressions, gestures, and the kiosk screen’s UI layout. P5 mentioned, *“Given that the AI banker’s kiosk combines a touchscreen with a screen for voice interaction, it was imperative for us to consider these elements. I believe it was a necessary approach for us.”* Participants in the project emphasized that these diverse elements needed to be a primary focus in the AI interaction guidelines.

4.3 Understanding the Dynamics within the Company

The final key insight from our interviews was the critical role of internal company dynamics in shaping AI interaction guidelines (P3, P4, P6, P9, P13). The project involved diverse groups including UX researchers, designers, and financial experts, who all concurred that the collaborative experiences significantly contributed to the

project, reflecting the unique dynamics of the company in the output. P5 shared, *“During this project, I particularly valued the time for active communication with industry professionals and designers, and engaging in healthy debates to ponder what better directions we could take.”*

Initially, the bank’s practitioners actively voiced the unique characteristics and CX values of the bank. They consistently shared their insights throughout the project to prevent the AI design guidelines from becoming too general or too abstract. P1 stated, *“At the project’s initiation, we had several meetings to share information about our AI projects and discuss what kind of guidance we wanted. As the project progressed, we provided feedback on how the guidelines could be practically applied after the framework was established.”* P3 added, *“We developed a practical draft for the interaction guide tailored to the finance sector. It was crucial to consider how these guidelines would apply in real banking operations and our channels.”*

The design agency played a substantial role in enhancing the practicality of the guidelines (P4, P5, P7). Our designers argued that AI interaction guidelines should go beyond listing principles and worksheets. P7 said, *“As a designer, I was responsible for creating many visual elements.”* They advocated for guidelines like Google Material Design, which details intricate aspects such as the size and placement of design components. They emphasized the necessity of developing these guidelines into a high-fidelity design system for actual design processes. P6 stated, *“Instead of just talking about general principles, it’s crucial to create UI design elements that can be directly applied in design.”*

The university’s UX research team played an essential role in areas that might be overlooked by bank employees or design agencies. P10 explained, *“We referred to existing AI design guidelines, conducted comprehensive analyses, and looked into relevant literature and research papers. We also analyzed cases from star services and other companies handling AI, focusing on this project.”* They conducted heuristic evaluations of the final output, offering concrete suggestions for improvement. P13 said, *“Through heuristic evaluations, we could reassess the final guidelines from an expert perspective, which helped us refine them.”* As Korean companies often collaborate with academia, the input from these external researchers was crucial, as the AI interaction guidelines are likely to be utilized not only by bank employees but also by these specialists.

Overall, understanding the dynamics among stakeholders related to interaction guidelines is vital in making these guidelines more practical and broadly applicable. Participants reported that collaboration among stakeholders was instrumental in the project’s success. The participants expected that the developed guidelines would serve as a basis for communication across departments, and hoped these tools would be adopted beyond the UX field within the bank. Particularly for bank UX practitioners, who frequently interact with data scientists and developers, these guidelines were seen as potential boundary objects for such collaborations.

5 DISCUSSION

Our research has revealed critical insights into the development of AI interaction guidelines within a specific domain. These findings underscore the multifaceted nature of such projects and highlight the necessity for guidelines that are not only functional, but also

resonate with the company's ethos, specific domain requirements, and the evolving nature of AI technology. We suggest considerations for AI design guidelines tailored to each domain's distinctive environment.

- **Incorporating Unique Company Philosophy, Legacy, and Values:** Our study reveals that AI guidelines must transcend technical specifications to encapsulate the core philosophy and values of the company. This involves understanding and integrating the company's vision, CX strategy, and ethical standards. For instance, in the financial domain, trustworthiness, security, and accuracy are paramount, reflecting the sector's conservative standards. Thus, guidelines must be tailored to reflect these fundamental values while ensuring AI interactions align with the company's legacy and ethos.
- **Creating Specialized Guidelines for Various Professional Fields:** The diversity of AI applications across different sectors necessitates domain-specific guidelines. For example, in finance, AI interactions must consider stringent security measures and regulatory compliance; while in healthcare, patient confidentiality and empathy are crucial. Therefore, guidelines must be uniquely developed for each professional field, considering their distinct challenges and nuances. This specialization not only enhances the relevance and effectiveness of AI interactions, but also ensures that domain-specific ethical considerations are adequately addressed.
- **Designing Guidelines for Continuous Growth:** AI technology is continuously evolving, with new developments like generative AI and conversational agents reshaping user expectations and interaction paradigms. Therefore, AI interaction guidelines should be dynamic, allowing for continuous growth and adaptation. This involves establishing a framework that can evolve with technological advancements and changing user needs. The guidelines should facilitate ongoing learning, encourage experimentation, and be open to revision, ensuring they remain relevant and effective over time. This approach also means that guidelines must be periodically reviewed and updated to reflect the latest AI advancements and industry best practices.

6 LIMITATIONS AND FUTURE WORK

Our research makes a significant contribution by suggesting strategies for making AI interaction guidelines more domain-specific and suitable for varied and dynamic work environments. However, there are still some limitations. First, since our study is based on a single case study, generalizing our findings to all UX work environments may be challenging. In future research, we intend to explore how AI interaction guidelines are designed in other fields (such as health, law, entertainment, etc.) by studying more examples. We also have not investigated the actual impact of the AI interaction guidelines on company UX practices, which was the subject of our study. This is partly because these guidelines have only recently been developed and are not yet actively used in practice. In future studies, we aim to examine how these guidelines are utilized by

UX practitioners in designing various financial AI services and any additional considerations that may be needed.

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